

Two-channel phenotype map of apparent generalization inflation

		Spatial-neighborhood channel	Patient-associated channel	Pattern
DLPFC	PCA+Ridge	buffered vs random 0.328	patient-held-out vs random 0.213	Mixed
	Spatial kNN	0.402	0.120	Spatial
Andersson multi-patient	PCA+Ridge	0.352	0.662	Patient-associated
	GraphSAGE	0.075	0.695	Patient-associated
Thrane multi-patient	PCA+Ridge	<0	0.499	Patient-associated
	GraphSAGE	NA	0.711	Patient-associated
Visium breast single patient	PCA+Ridge	0.259	NA	Spatial
	Spatial kNN	0.796	NA	Spatial
	GraphSAGE	0.265	NA	Spatial
GSE278936 spatial replication	PCA+Ridge	0.222	NA	Spatial replication
	Spatial kNN	NA	NA	Boundary

Cell values are RLI; stronger shading indicates larger positive attenuation. Hatched NA is unavailable or non-interpretable; <0 is negative/no positive inflation.